

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flask Loan Prediction (/predict)
Test Environment	Local Machine (Windows 11, Python 3.11, Flask)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario	No. of Virtual Users	Duration	Expected Outcome
1	Single prediction request	1	60 sec	Prediction generated successfully
2	Multiple simultaneous prediction requests	10	120 sec	All requests processed successfully
3	Continuous prediction requests	25	180 sec	Stable response without errors

4	Peak load test	50	300 sec	Application remains responsive with acceptable response time
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Step 3: Performance Test Results

Metric	Target Value	Actual Value	Status	Remarks	Metric
Average Response Time	<2 sec	0.84 sec	Pass	Prediction generated quickly	Average Response Time
Maximum Response Time	<5 sec	2.31 sec	Pass	Within acceptable limit	Maximum Response Time
Throughput	—	18 requests/sec	Pass	Stable request handling	Throughput
Error Rate	<1%	0%	Pass	No failed requests	Error Rate
CPU Utilization	<80%	38%	Pass	CPU usage remained low	CPU Utilization
Memory Utilization	<80%	46%	Pass	Stable memory usage	Memory Utilization

Step 4: Observations & Analysis

- The Flask application successfully handled concurrent prediction requests without failures.
- Average response time remained below the target threshold of 2 seconds.
- CPU and memory utilization stayed within acceptable limits throughout testing.
- No request failures or prediction errors were observed during the load test.
- The machine learning model loaded efficiently and generated predictions consistently.
- The application is suitable for deployment in small to medium-scale environments and can be further optimized for higher user loads through cloud scaling and load balancing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

